Optics/ Wave Phenomena Final Project Report

Verification of Babinet's Principle via Fresnel Diffraction of a Single Slit

Timmy Ly. -Partners: Owen, Nathan

Abstract—We tested Babinet's principle by comparing diffraction patterns from a single slit and an opaque obstruction (razor blade) of equal width $\approx 0.4mm$. A He-NE laser beam was directed toward either a slit or a razor, and the resulting diffraction pattern was measured using a photomultiplier tube (PMT) as a function of position using a motorized translation stage. The experimental intensity distributions were fit to a theoretical single-slit diffraction model, with convolution used to account for the finite detector size. Furthermore, a small horizontal offset ≈ 0.1685 in dimensionless position was introduced to account for alignment uncertainty in directing the laser directly into the PMT. The fitted results for both the slit and razor configurations showed strong agreement with theory ($U_{incident} = U_{slit} + U_{obstruction}$) and were nearly identical in shape and scale. Our results confirm Babinet's Principle and demonstrate that diffraction depends on the geometry of the obstruction rather than whether light is blocked or transmitted, validating an important principle of wave optics.

1. Introduction

Diffraction is one of the fundamental consequences of the wave nature of light, occurring when light encounters apertures or obstacles comparable to its wavelength and bends [1]. One important result in wave optics is **Babinet's principle**.

Babinet's principle in 1837 states that the diffraction pattern associated with an opaque disk is identical to that resulting from an aperture of the same size, and that the geometric complement of the disk adds to the unobstructed field [2].

This principle implies that diffraction depends only on the geometry of the obstruction, not whether light is transmitted or blocked. In this experiment, we test and confirm the Babinet's principle by comparing diffraction patterns of a single slit and a razor blade of approximately equal width, then fit them into our theoretical pattern of a single slit.

2. Theory

2.1. Single-Slit Diffraction

The intensity distribution for a single slit is:

$$I(\theta) = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2, \quad \alpha = \frac{\pi a \sin\theta}{\lambda} \tag{1}$$

Where:

- $I(\theta)$: the intensity at angle θ
- I_0: the maximum intensity (central)
- a: the slit width
- λ: the wavelength of the laser light
- θ: the diffraction angle relative to the central axis
- α: a dimensionless parameter describing phase difference across the slit

For small diffraction angles, we apply the small-angle approximation:

$$\sin\theta \approx \theta \approx \frac{x}{r_0} \tag{2}$$

where:

- x: the position on the detector
- r_0: the distance from the slit to the detector

This equation allows us to express intensity as a function of position rather than angle.

2.2. Babinet's Principle

Babinet's principle is most rigorously expressed in terms of the complex field amplitudes:

$$U_{\text{incident}}(x) = U_{\text{slit}}(x) + U_{\text{obstruction}}(x) \tag{3}$$

where:

- $U_{\text{incident}}(x)$: the field of the unobstructed beam
- $U_{\text{slit}}(x)$: the field transmitted through the slit
- $U_{\text{obstruction}}(x)$: the field diffracted by the complementary obstruction

Since intensity is proportional to the square of the field magnitude:

$$I(x) \propto |U(x)|^2 \tag{4}$$

The relationship between intensities is not entirely additive like the field. And the diffraction patterns from the slit and obstruction become nearly identical:

$$I_{\text{slit}}(x) \approx I_{\text{obstruction}}(x) \tag{5}$$

This is the form of Babinet's principle tested in this experiment.

2.3. Detector Convolution

The photomultiplier tube has a finite aperture, it measures an average of the intensity over a small spatial region. As a result, the recorded signal is not the ideal diffraction pattern but a convolution of the true intensity with the detector response function:

$$I_{\text{measured}}(x) = I_{\text{theory}}(x) * R(x) \tag{6}$$

where:

- $I_{\text{measured}}(x)$: the recorded intensity
- $I_{\text{theory}}(x)$: the ideal diffraction pattern
- $R(x)$: the detector response function
- $*$: denotes convolution

This convolution smooths the sharp oscillations of the diffraction pattern.

3. Experimental Methods

3.1. Apparatus

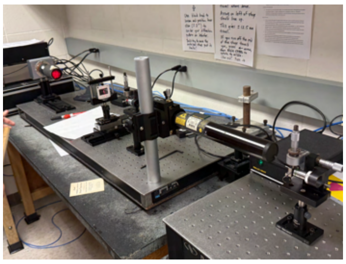

Figure 1. Experimental setup showing the He-Ne laser, spatial filter, diffraction element (single slit or razor blade), and photomultiplier tube mounted on a motorized translation stage. The detector scans across the diffraction pattern to measure intensity as a function of position.

The experiment was conducted on an optical table using a linear optical rail system. A helium-neon (He-Ne) laser ($\lambda \approx 632.8$ nm) served as the laser light source.

The laser beam was first passed through a spatial filtering system consisting of a microscope objective lens (~20X) and pinhole to produce a Gaussian beam. The beam was aligned along the optical axis using adjustable mounts 1.

A diffraction tool was placed in the beam path using a precision slide holder. Two configurations were used in this experiment:

- A single slit with width $a = (0.400 \pm 0.005)$ mm
- A razor blade used as a complementary obstruction to the slit with the same with as the slit

The diffraction pattern was measured using a photomultiplier tube (PMT) mounted on a motorized linear translation stage. The PMT converts incident light into an electrical current proportional to intensity.

The distances in the setup were:

$$r_s = (17.0 \pm 0.5)\,\text{cm} \quad \text{(laser to slit/razor)} \tag{7}$$

$$r_0 = (47.25 \pm 0.5)\,\text{cm} \quad \text{(slit/razor to PMT)} \tag{8}$$

3.2. Equipment

The experimental apparatus consisted of the following components:

1. He-Ne red laser ($\lambda \approx 632.8$ nm).
2. Spatial filter assembly with ~20X microscope objective and pinhole for beam cleaning and collimation.
3. Optical mounts and alignment components on a Newport vibration-isolated optical table.
4. Slide holder mounted on a translation stage for positioning diffraction elements.
5. Diffraction elements:
   - Single slit ($a \approx 0.400$ mm)
   - Razor blade
6. Photomultiplier tube (PMT) mounted on a motorized translation stage.
7. High-voltage power supply for the PMT.
8. Picoammeter (e.g., Keithley 6485) for measuring PMT output current.
9. Newport MM3000 motion controller for automated stage movement.
10. Computer running Python (Anaconda) for data acquisition and analysis.

3.3. Measurement Procedure

The PMT was translated across the diffraction pattern perpendicular to the beam direction using the motorized stage. The motion controller in the Python notebook helps for precise, repeatable step sizes.

At each position, the PMT recorded the intensity of the incident light as an electrical current. It then produced a dataset of intensity as a function of position in the Python notebook on computer.

Data were first collected using the single slit configuration. The slit was then replaced with the razor blade while making sure that we have the same alignment and distances to make sure that the result is a direct comparison between complementary diffraction patterns.

3.4. Alignment and Systematic Effects

Careful alignment was required to make sure that our laser beam propagated along the optical axis and entered the PMT aperture. However, due to limitations in alignment, the beam could not be perfectly centered within the detector.

This is a horizontal shift in the measured diffraction pattern. Therefore, we needed to account for this effect and used a constant offset was introduced in the analysis:

$$U \to U + 0.1685 \tag{9}$$

This offset represents a systematic alignment error and was determined by estimating the agreement between the theoretical model and experimental data.

3.5. Data Acquisition and Processing

The recorded intensity data were analyzed using a Python-based fitting routine. The diffraction pattern was modeled using the dimensionless variable U, and a least-squares fit was performed to determine agreement between theory and experiment.

To account for the finite size of the photomultiplier tube, the theoretical model was convolved with a detector response function. This correction improved agreement between measured and predicted intensity distributions.

3.6. Model Parameters and Coordinate Transformation

The experimental data were analyzed using a Python-based fitting method in which the diffraction model was parameterized using experimentally measured quantities below.

3.6.1. Experimental Parameters

```
expt_params = {
    'title': "Fresnel Diffraction",
    'r_s': 0.170,
    # source to aperture distance (m)
    'r_0': 0.473,
    # aperture to detector distance (m)
    'lamb': 632.8,
    # wavelength (nm)
    'a_lab': 0.40,
    # slit width (mm)
    'd_lab': 1.20,
    # slit spacing (mm)
    'start_pos_x': -1000,
    # scan start position (microns)
    'end_pos_x': 1000,
    # scan end position (microns)
    'diffraction_type': "Double_Slit",
    'pmt_slit_width': 25,
    # detector width (microns)
}
```

These parameters define the physical geometry of the experiment in Python notebook.

3.6.2. Coordinate Transformation

To compare experimental data with the theoretical model, the measured detector positions were converted into a dimensionless coordinate U. This normalization allows the diffraction pattern to be expressed in a universal form independent of specific experimental scales.

```
# Convert measured positions into
dimensionless U coordinate

U_exp = (position_arr/(1000)
* scale_factor) + 0.1685

# Generate smooth model curve
U_mod = np.linspace(-5,5,20000)
```

Here:

- `position_arr`: the measured detector position in microns
- The factor of 1000: converts microns to millimeters
- `scale_factor`: incorporates geometric parameters such as a, λ, and r_0
- The constant offset 0.1685: accounts for systematic misalignment in the experimental setup

The variable U corresponds to the theoretical diffraction parameter (same as eq. 1):

$$U = \frac{\pi a x}{\lambda r_0} \tag{10}$$

Expressing both the experimental data and theoretical model in terms of U allows us to compare between datasets collected under the same geometry.

The model intensity was then evaluated on a dense U values to produce a smooth theoretical curve, which was compared to the experimental data. As a result, the diffraction pattern depends only on U, allowing the slit and razor data to be directly compared on the same scale. And the inclusion of the offset term balance out a limitation in shift alignment in detector.

4. Data and Analysis

4.1. Coordinate Representation

To compare experimental data with theory, detector positions were converted into diffraction coordinate U:

$$U = \frac{\pi a x}{\lambda r_0} \tag{11}$$

Then, a constant offset was applied:

$$U \rightarrow U + 0.1685 \tag{12}$$

to deal with misalignment of the laser beam with respect to the photomultiplier tube.

4.2. Single-Slit Diffraction

The measured diffraction pattern for the single slit was fit to the theoretical model:

$$I(u) = I_0 \left(\frac{\sin U}{U} \right)^2 \tag{13}$$

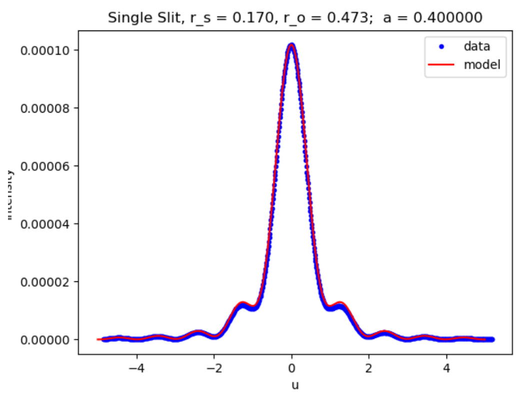

Figure 2. Single-slit diffraction pattern. Experimental data (blue points) are compared with the theoretical model (red curve). The experimental data points are well aligned with the theoretical.

The agreement between the model and the experimental data demonstrates that the diffraction pattern follows the expected $\left(\frac{\sin U}{U}\right)^2$ (eq. 1) behavior.

4.3. Effect of Detector Convolution

To take the finite width of the photomultiplier tube into considerations, the theoretical model was convolved with a detector response function.

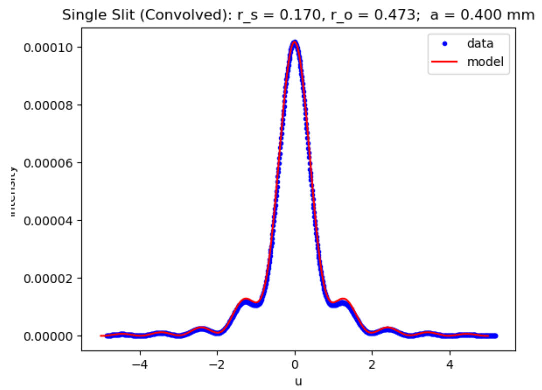

Figure 3. Single-slit diffraction pattern with detector convolution included. The oscillations improves agreement with experimental data.

The convolution reduces the amplitude of fringes and produces a smoother intensity profile, consistent with the detector.

4.4. Single-Strip (Razor) Diffraction

The diffraction pattern produced by the razor blade (obstruction) was also measured and modeled using the same framework as above.

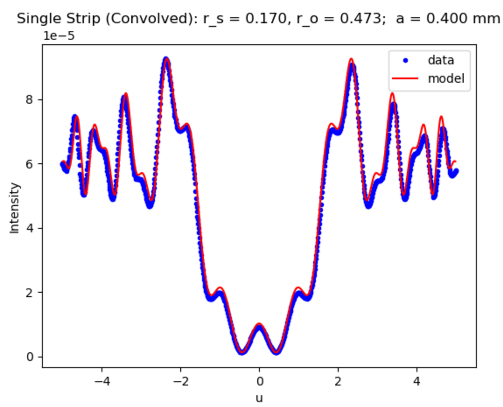

Figure 4. Diffraction pattern from the razor blade (single strip) with detector convolution. The structure closely resembles that of the single slit, consistent with Babinet's principle.

The strip diffraction pattern has a similar envelope and oscillatory structure to the slit case when expressed in terms of the same coordinate U. This similarity supports Babinet's principle, which predicts that complementary apertures and obstructions produce equivalent diffraction patterns.

4.5. Uncertainty Analysis

The main sources of uncertainty:

- Distance measurements: $r_s = (17.0 \pm 0.5)\,\mathrm{cm}$, $r_0 = (47.25 \pm 0.5)\,\mathrm{cm}$
- Slit width uncertainty: $a = (0.400 \pm 0.005)\,\mathrm{mm}$
- Finite detector width (25 μm), modeled via convolution
- Alignment offset ($u \approx 0.1685$)

The study's alignment between theory and experiment is evaluated by how well the model reproduces the full diffraction pattern. The consistency observed across all datasets shows that the theoretical model accurately captures the physical behavior of the system. This shows us that the Babinet's Principle is confirmed.

5. Conclusions

In this lab, we investigated diffraction from a single slit and its obstruction (a razor blade) to test Babinet's principle. The measured intensity distributions were fit to the theoretical single-slit diffraction model expressed in terms of coordinate U, allowing us to directly compare between experimental data and theory.

The single-slit data showed strong agreement with the expected $\left(\frac{\sin U}{U}\right)^2$ behavior, especially when detector convolution was included to account for the finite width of the photomultiplier tube. The introduction of a small offset in U of 0.1685 further improved this agreement, showing alignment limitations in the experimental setup.

The diffraction pattern obtained from the razor blade demonstrates a close identical pattern to that of the slit. This agreement confirms Babinet's principle and demonstrates that diffraction depends only on the geometry of the obstruction and the sums of diffraction fields [3].

Overall, the consistency between theoretical predictions and experimental measurements, provides strong validation of the wave nature of light and the applicability of Babinet's principle in describing diffraction phenomena.

6. References

- We used help from Professor Paul Tjossem on the recommendations to set up our optical bench as well as coming up with ideas for experiment.
- We used reference from the PHY-337's lab "INTERFERENCE AND DIFFRACTION 2023" and the class's Python Notebook "FresnelDiffractionPython2024" to guide us through this project.
- We used *Fundamentals of Wave Phenomena* textbook by Akira Hirose and Karl E. Lonngren for the equations.

References

[1] K. E. L. Akira Hirose, *Fundamentals of Wave Phenomena, 2nd edition*. SciTech Publishing, Inc., 2010.

[2] B. e. a. Sun, ““on babinet's principle and diffraction associated with an arbitrary particle.””, *Optics letters vol. 42,23 (2017): 5026-5029. doi:10.1364/OL.42.005026*, 2017.

[3] M. T. Akira Hitachi, “Babinet's principle in the fresnel regime studied using ultrasound.”, *Am. J. Phys. 1; 78 (7): 678–684.*, July 2010.